



SURYA S

Mechatronics Engineer | Industrial Automation | AI Computer Vision | IIoT | Product Development

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ABOUT ME

Mechatronics Engineer with 6+ years of industrial experience in electrical and mechanical maintenance, Industrial Automation, AI Computer Vision, IIoT, and Product Development. Successfully developed and **deployed 25+ industrial automation and digitalization projects focused on improving productivity**, predictive maintenance, process optimization, industrial safety, and operational efficiency. Experienced in delivering complete engineering solutions by integrating Mechanical Design, PLCs, Embedded Systems, Python, Flask, OpenCV, and custom YOLO models from concept to deployment.

EDUCATION

SWISS VOCATIONAL DIPLOMA IN MECHATRONICS Texmo Academy (SkillSonics) 2017- 2020

Completed a Swiss vocational training program specializing in Mechanical Engineering, Electrical Systems, Industrial Automation, PLCs, Manufacturing, Pneumatics, and Product Development.

SKILLS

PROGRAMMING & AI	INDUSTRIAL AUTOMATION	EMBEDDED SYSTEMS
•Python •Flask •OpenCV •YOLO •HTML •CSS •JavaScript •SQL	•Delta PLC & HMI •Siemens PLC •Allen Bradley •Mitsubishi •ABB •CE Fanuc •Omron •SCADA •HMI •FactoryTalk •WinCC •Intouch	•Raspberry Pi •Arduino •ESP32 •Industrial Sensors •RS232 •RS485 •Camera Integration •Serial Communication
MECHANICAL DESIGN	INDUSTRIAL ENGINEERING	
Solidworks • 3D Printing • Machine Design ReverseEngineering`•Prototype Development Assembly • Manufacturing	• TPM • Kaizen • SQDIP • Preventive Maintenance • Root Cause Analysis (RCA) • A3 Problem Solving	

EXPERIENCE

JUNIOR ENGINEER – ELECTRICAL, MECHANICAL & INDUSTRIAL AUTOMATION Texmo Industries, Coinbatore May 2020 – Present

- Spearheaded the digital transformation of maintenance operations by **developing 25+ Industrial Automation, IIoT, AI, and Industry 4.0 solutions, improving productivity, troubleshooting efficiency, and operational visibility.**
- Designed and deployed web-based applications including Energy Monitoring System (**EMS**), Overall Equipment Effectiveness (**OEE**), Condition-Based Monitoring (**CBM**), and Time-Based Monitoring (**TBM**) to support predictive maintenance and data-driven decision making
- Developed an AI-powered Fabric **Defect Detection System** using Python, Flask, OpenCV, and custom-trained **YOLO models**, integrating 4 industrial cameras, live video streaming, automated defect detection, and digital logging.
- Built **Computer Vision-based Industrial Safety Monitoring Systems**, including Line Crossing Detection and real-time safety alert applications.
- Developed a **Digital Engineering Management System** that enabled maintenance technicians to instantly access machine drawings, identify spare parts, and **reduce troubleshooting time.**
- Created a **Breakdown Assistant Web Application** with troubleshooting procedures, images, and root cause references, enabling faster fault diagnosis, standardized maintenance practices, and **reduced machine downtime.**
- Designed and implemented a **Barcode-based Spare Inventory & Consumption Tracking System with real-time availability, usage history, and automated notifications to improve maintenance planning.**
- Integrated PLCs, HMIs, Raspberry Pi, ESP32, Arduino, industrial sensors, and web technologies to build complete end-to-end industrial automation solutions.
- Performed **Preventive Maintenance, Root Cause Analysis (RCA), TPM, A3 Problem Solving, Continuous Improvement, and Kaizen initiatives** to improve equipment reliability and manufacturing efficiency.
- **Awarded First Prize in the Continuous Improvement** Competition for implementing an innovative A3 Problem Solving project and actively contributed to **20+ Kaizen initiatives.**